Question	Answer	Marks	Partial Marks
6(a)	52[.0] or 52.02…	**4**	**M2** for [cos =] $\frac{13^2+4^2-11^2}{2\times13\times4}$ or **M1** for $11^2 = 13^2+4^2-2\times13\times4\cos(...)$ **A1** for [$\cos^{-1}$ =] $\frac{64}{104}$ oe or 0.615 or 0.6153 to 0.6154
6(b)	62.7 or 62.69 to 62.70	**4**	**M3** for $180 - \sin^{-1}\left(\frac{8\sin 80}{13}\right) - 80$ oe or **M2** for $\sin A = \frac{8\sin 80}{13}$ or **M1** for $\frac{13}{\sin 80} = \frac{8}{\sin A}$ oe **A1** for 37.3 or 37.30… If 0 scored, **M1** for 180 – 80 – *their A*
6(c)	66.7 or 66.68 to 66.71	**3**	**M1** for $0.5\times13\times4\times\sin(theirACB)$ oe **M1** for $0.5\times8\times13\times\sin(their\ ACD)$ oe
7(a)(i)	$\frac{3}{5} > \frac{1}{4}$ oe or $\frac{12k}{20k}$ and $\frac{5k}{20k}$ or 0.6 and 0.25 or 60% and 25%	**1**	
7(a)(ii)	$\frac{11}{20}$ oe	**3**	**M2** for $\frac{3}{5}\times\frac{3}{4} + \frac{2}{5}\times\frac{1}{4}$ oe or $1 - \frac{3}{5}\times\frac{1}{4} - \frac{2}{5}\times\frac{3}{4}$ oe or **M1** for $\frac{3}{5}\times\frac{3}{4}$ or $\frac{2}{5}\times\frac{1}{4}$ oe (but not as part of a larger product)
7(b)(i)	$\frac{6}{60}$ oe	**2**	**M1** for $\frac{3}{5}\times\frac{2}{4}\times\frac{1}{3}$ oe If 0 scored, **SC1** for answer $\frac{27}{125}$ oe
7(b)(ii)	0	**1**	Accept $\frac{0}{60}$

Question	Answer	Marks	Partial Marks
7(c)	$\frac{11}{25}$ oe	**3**	**M2** for $\frac{3}{5}\times\frac{3}{5}+\frac{2}{5}\times\frac{1}{5}$ oe or $1-\frac{3}{5}\times\frac{2}{5}-\frac{2}{5}\times\frac{4}{5}$ oe or **M1** for $\frac{3}{5}\times\frac{3}{5}$ or $\frac{2}{5}\times\frac{1}{5}$ or for a correct tree showing all 25 outcomes with the 11 correct outcomes identified
8(a)(i)	4	**2**	**M1** for correct method using similar triangles e.g. $\frac{10}{5}=\frac{8}{DX}$ oe
8(a)(ii)	36.9 or 36.86 to 36.87	**2**	**M1** for tan = $\frac{6}{8}$ or sin = $\frac{6}{10}$ or cos = $\frac{8}{10}$ oe
8(b)	[v =] 150	**B1**	
	[w =] 15	**B2**	**FT** (180 – *their v*) ÷ 2 **M1** for 180 – 2w = *their v* oe or angle POQ = 180 – *their v* oe
	[x =] 15	**B1**	**FT** *their w*
	[y =] 10	**B2**	**M1** for angle TPS = 5° or angle TXS = 20° or OXP = 20° or TXP = 160° (where X is where OT and PS intersect)
8(c)	182 or 182.4…	**3**	**M2** for $\left(\frac{94}{226}\right)^{\frac{3}{2}}\left[=\frac{V}{680}\right]$ oe or **M1** for ratio of lengths = $\sqrt{\frac{226}{94}}$ or $\sqrt{\frac{94}{226}}$ or better or for $\frac{V^2}{680^2}=\frac{94^3}{226^3}$ oe
9(a)	0	**1**	
9(b)	5	**2**	**M1** for $3(3^x)+4$ or better or f($\frac{1}{3}$) or f(3^{-1})
9(c)	$\frac{x+1}{2}$ oe final answer	**2**	**M1** for $x=2y-1$ or $y+1=2x$ or $\frac{y}{2}=x-\frac{1}{2}$ or better
9(d)	$9x+16$	**2**	**M1** for 3(3x + 4) + 4 oe
9(e)	$9x^2+24x+16$	**2**	**B1** for three terms from $9x^2+12x+12x+16$ correct
9(f)	27	**2**	**M1** for x = h(*their* g(2))